Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If a magnet is suspended at an angle of 30° to the magnetic meridian, the dip needle makes an angle of 45° with the horizontal. The real dip is

(A) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(B) $\tan^{-1}(\sqrt{3})$

(C) $\tan^{-1}\left(\sqrt{\frac{3}{2}}\right)$

(D) $\tan^{-1}\frac{2}{\sqrt{3}}$

Q.2

If the period of oscillation of freely suspended bar magnet in earth's horizontal field H is $4\mathrm{sec}$. When another magnet is brought near it, the period of oscillation is reduced to 2 s. The magnetic field of second bar magnet is

(A) $4\mathrm{H}$

(B) $3\mathrm{H}$

(C) $2\mathrm{H}$

(D) $\sqrt{3}\mathrm{H}$

Q.3

A compass needle whose magnetic moment is $60\mathrm{Am}^2$, is directed towards geographical north at any place experiencing moment of force of $1.2 \times 10^{-3}\mathrm{Nm}$. At that place the horizontal component of earth field is 40 micro $\mathrm{W/m}^2$. What is the value of dip angle at that place?

(A) 30°

(B) 60°

(C) 45°

(D) 15°

Q.4

The angle of dip at a place is 37° and the vertical component of the earth's magnetic field is 6×10^{-5} T. The earth's magnetic field at this place is $(\tan 37^{\circ} = 3/4)$

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(A) $7 \times 10^{-5}\ \mathrm{T}$

(B) $6 \times 10^{-5}\ \mathrm{T}$

(C) $5 \times 10^{-5}\ \mathrm{T}$

(D) $10^{-4}\ \mathrm{T}$

Q.5

A dip circle is so set that its needle moves freely in the magnetic meridian. In this position, the angle of dip is 40°. Now the dip circle is rotated so that the plane in which the needle moves makes an angle of 30° with the magnetic meridian. In this position, the needle will dip by an angle

(A) 40°

(B) 30°

(C) more than 40°

(D) less than 40°

Q.6

The earth's magnetic field lines resemble that of a dipole at the centre of the earth. If the magnetic moment of this dipole is close to $8 \times 10^{22} \mathrm{Am}^2$, the value of earth's magnetic field near the equator is close to (radius of the earth $= 6.4 \times 10^6\ \mathrm{m}$)

(A) 0.6 Gauss

(B) 1.2 Gauss

(C) 1.8 Gauss

(D) 0.32 Gauss

Q.7

The magnetic field of earth at the equator is approximately 4 $\times 10^{-5}\ \mathrm{T}$. The radius of earth is $6.4 \times 10^6\ \mathrm{m}$. Then what is the dipole moment (in Am^2) of the earth? (Enter the number which is power of 10 in the answer expression)

(A) 25

(B) 26

(C) 22

(D) 23

Q.8

The strength of the earth's magnetic field is

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(A) constant everywhere

(B) zero everywhere

(C) having very high value

(D) vary from place to place on the earth's surface

Q.9

Which of the following is responsible for the earth's magnetic field

(A) Convective currents in earth's core.

(B) Diversive current in earth's core.

(C) Rotational motion of earth.

(D) Translational motion of earth.

Q.10

One can define $\ldots A \ldots$ of a place as the vertical plane which passes through the imaginary line joining the magnetic North and the South poles. Here, A refers to

(A) geographic meridian

(B) magnetic meridian

(C) magnetic declination

(D) magnetic inclination

Q.11

The earth's magnetic field lines resemble that of
a dipole at the centre of the earth. If the magnetic moment of this dipole is close to $8 \times 10^{22} \mathrm{Am}^2$, the value of earth's magnetic field near the equator is close to (radius of the earth $= 6.4 \times 10^6$ m)

(A) 0.6 Gauss

(B) 1.2 Gauss

(C) 1.8 Gauss

(D) 0.32 Gauss

Q.12

The magnetic field of earth at the equator is approximately 4×10^{-5} T. The radius of earth is 6.4×10^6 m . Then the dipole moment of the earth will be nearly of the order of

(A) 10^{23} A m^2

(B) 10^{20} A m^2

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(C) 10^{16} A m^2

(D) $10^{10} Am^2$

Q.13

The true value of angle of dip at a place is 60°, the apparent dip in a plane inclined at an angle of 30° with magnetic meridian is

(A) $tan^{-1}\frac{1}{2}$

(B) $tan^{-1}(2)$

(C) $tan^{-1}\left(\frac{2}{3}\right)$

(D) $tan^{-1}\left(\frac{\sqrt{3}}{4}\right)$

Q.14

Let V and H be the vertical and horizontal components of earth's magnetic field at any point on earth. Near the North pole

(A) $\mathrm{V} >> \mathrm{H}$

(B) $\mathrm{V} << \mathrm{H}$

(C) $\mathrm{V} = \mathrm{H}$

(D) $\mathrm{V} = \mathrm{H} = 0$

Q.15

A torque of $10^{-5}\mathrm{Nm}$ is required to hold a magnet at 90° with the horizontal component H of the earth's magnetic field. The torque to hold it at 30° will be

(A) $5 \times 10^{-6}\mathrm{Nm}$

(B) $5\mathrm{Nm}$

(C) $\frac{1}{3} \times 10^{-5}\mathrm{Nm}$

(D) $5\sqrt{3} \times 10^{-6}\mathrm{Nm}$

Answers & Solutions

Q.1 Answer:

$\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

Solution:

$\tan\delta = \tan\delta' \cos\alpha = \tan 45^\circ \cos 30^\circ$

$\delta = \tan^{-1}\frac{\sqrt{3}}{2}$

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(a)

Q.2 Answer:

$4\mathrm{H}$

Solution:

The time period of oscillation of a freely suspended magnet is given by,

$$\mathrm{T}=2\pi\sqrt{\frac{\mathrm{I}}{\mathrm{MH}}}$$

$$\text{Thus,}\quad \frac{\mathrm{T}}{\mathrm{T}'}=\frac{2\pi\sqrt{\frac{\mathrm{I}}{\mathrm{MH}}}}{2\pi\sqrt{\frac{\mathrm{I}}{\mathrm{MH}'}}}$$

Given, $\mathrm{T}=4\mathrm{sec}, \mathrm{T}'=2\mathrm{sec}$

So, $\frac{4}{2}=\sqrt{\frac{\mathrm{H}'}{\mathrm{H}}}$

or, $\sqrt{\frac{\mathrm{H}'}{\mathrm{H}}}=2$

or, $\mathrm{H}'=4\mathrm{H}$

Q.3 Answer:

30°

Solution:

$$\mathrm{M}=60\mathrm{Am}^2$$

$$\vec{\tau}=1.2\times10^{-3}\mathrm{Nm}, \mathrm{B_H}=40\times10^{-6}\ \mathrm{Wb/m^2}$$

$$\vec{\tau}=\overrightarrow{\mathrm{M}}\times\overrightarrow{\mathrm{B}}_{\mathrm{H}}\Rightarrow\tau=\mathrm{MB_H}\sin\theta$$

$$\Rightarrow 1.2\times10^{-3}=60\times40\times10^{-6}\sin\theta$$

$$\Rightarrow\sin\theta=\frac{1.2\times10^{-3}}{60\times40\times10^{-6}}=\frac{1}{2}=\sin30^{\circ}$$

$$\Rightarrow\theta=30^{\circ}$$

Q.4 Answer:

$10^{-4}\ \mathrm{T}$

Solution:

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$$\tan\delta = \frac{V}{H} = \frac{3}{4}\left[\because \tan 37^\circ = \frac{3}{4}\right]$$

$$\therefore \quad V = \frac{3}{4}H$$

$$V = 6 \times 10^{-5}\ T$$

$$H = \frac{4}{3} \times 6 \times 10^{-5}\ T = 8 \times 10^{-5}\ T$$

$$\therefore B_{total} = \sqrt{V^2 + H^2} = \sqrt{(36+64)} \times 10^{-5}$$

$$= 10 \times 10^{-5} = 10^{-4}\ T.$$

Q.5 Answer:

less than 40°

Solution:

$= \sqrt{5}\frac{\mu_0}{4\pi}\frac{M}{d^3} = \sqrt{5}\,B, \{\, B_Q = \frac{\mu_0}{4\pi}\frac{M}{d^3} = B\}$

$\delta_1 = 40^\circ, \delta_2 = 30^\circ, \delta = ?$

$\cot\delta = \sqrt{\cot^2\delta_1 + \cot^2\delta_2} = \sqrt{\cot^2 40^\circ + \cot^2 30^\circ}$

$\cot\delta = \sqrt{1.19^2 + 3} = 2.1$

$\therefore \quad \delta = 25^\circ$ **i.e.** $\delta < 40^\circ$.

Q.6 Answer:

0.6 Gauss

Solution:

Given $M = 8 \times 10^{22} Am^2$

$d = R_e = 6.4 \times 10^6\ m$

Earth's magnetic field, $B = \frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3}$

$= \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{2 \times 8 \times 10^{22}}{(6.4 \times 10^6)^3}$

$\cong 0.6$ **Gauss**

Q.7 Answer:

23

Solution:

Given, $B = 4 \times 10^{-5}\ T$

$R_E = 6.4 \times 10^6\ m$

Dipole moment of the earth $M = ?$

$B = \frac{\mu_0}{4\pi}\frac{M}{d^3}$

$$4 \times 10^{-5} = \frac{4\pi \times 10^{-7} \times M}{4\pi \times (6.4 \times 10^6)^3}$$

$\therefore \quad M \cong 10^{23} Am^2$

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Q.8 Answer:

vary from place to place on the earth's surface

Solution:

The strength of the earths magnetic field is not constant. It varies from one place to other place on the surface of earth. Its value being of the order of 10^{-5} T.

Q.9 Answer:

Convective currents in earth's core.

Solution:

The earth's core is hot and molten. Hence, convective current in earth's core is responsible for it's magnetic field.

Q.10 Answer:

magnetic meridian

Solution:

No solution available

Q.11 Answer:

0.6 Gauss

Solution:

Given $\mathrm{M}=8 \times 10^{22} \mathrm{Am}^2$

$\mathrm{d}=\mathrm{R}_{\mathrm{e}}=6.4 \times 10^6 \mathrm{~m}$

Earth's magnetic field, $\mathrm{B}=\frac{\mu_0}{4 \pi} \cdot \frac{2 \mathrm{M}}{\mathrm{d}^3}$

$=\frac{4 \pi \times 10^{-7}}{4 \pi} \times \frac{2 \times 8 \times 10^{22}}{(6.4 \times 10^6)^3}$

$\simeq 0.6$ **Gauss**

Q.12 Answer:

10^{23} A m^2

Solution:

Given, $\mathrm{B}=4 \times 10^{-5} \mathrm{~T}$

$\mathrm{R}_{\mathrm{E}}=6.4 \times 10^6 \mathrm{~m}$

Dipole moment of the earth $\mathrm{M}=$ **?** $\mathrm{B}=\frac{\mu_0}{4 \pi} \frac{\mathrm{M}}{\mathrm{d}^3}$

$4 \times 10^{-5}=\frac{4 \pi \times 10^{-7} \times \mathrm{M}}{4 \pi \times(6.4 \times 10^6)^3}$

$\mathrm{M} \simeq 10^{23} \mathrm{Am}^2$

Q.13 Answer:

$tan^{-1}(2)$

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Solution:

$$\tan\theta' = \frac{\tan\theta}{\cos\alpha} = \frac{\tan 60^\circ}{\cos 30^\circ} = \frac{\sqrt{3}}{\sqrt{3}/2} = 2$$
$$\therefore \quad \theta' = \tan^{-1}(2)$$

Q.14 Answer:

$\mathrm{V} >> \mathrm{H}$

Solution:

Near the North pole $\mathrm{V} >> \mathrm{H}$ because there angle of dip is nearly 90° andhence $\mathrm{H} = \mathrm{B_E}\cos\delta$ is very very small and $\mathrm{V} = \mathrm{B_E}\sin\delta$ is nearly equal to $\mathrm{B_E}$.

Q.15 Answer:

$5 \times 10^{-6}\mathrm{Nm}$

Solution:

No solution available

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